

Younes AROUK

Final-Year Internship – DRE / Component Owner, Quality and Industrial Methods
Automotive and Digital Solutions – ENSA Berrechid
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PROFILE

Final-year engineering student in Automotive and Digital Solutions at ENSA Berrechid and Design Release Engineer (DRE) apprentice at SEGULA Technologies. I support product change tracking in PLM, risk analysis and preparation for component validation.

With a professional bachelor's degree in Metrology, Quality, Safety and Environment and experience in manufacturing methods and process diagnostics, I am seeking a final-year internship where I can help validate automotive components, improve product quality and use Lean methods to reduce waste and improve industrial processes.

PROFESSIONAL EXPERIENCE

DRE APPRENTICESHIP – PRODUCT CHANGES AND QUALITY
SEGULA Technologies Sep. 2026 – Present
- Track product changes, update technical documentation and maintain configuration traceability in PLM.
- Support risk analysis and validation planning in line with APQP and PPAP processes.

ENGINEERING INTERNSHIP – AUTOMOTIVE SYSTEM RELIABILITY
SEGULA Technologies Jul. – Aug. 2026
- Improved the reliability of ECU software updates on a simulated platform through automatic rollback and failure testing in QEMU.
- Improved monitoring of **28 ECUs simulated in Python** using a dashboard and log analysis.

MANUFACTURING METHODS INTERNSHIP – PRODUCTION PROCESSES
SWES – Berrechid Jun. 2026 – 1 month
- Analyzed automotive wiring harness manufacturing to identify opportunities for improvement and studied how customer changes were introduced into production.

INTERNSHIP – AFTER-SALES INVENTORY TRACKING
Auto Hall – Safi Jul. 2025 – 1 month
- Studied the after-sales workflow and the handoffs between the service desk, workshop and spare parts department.
- Developed a web application for tracking spare parts inventory. **Proposed targets:** centralize records for **100 part numbers** and cut lookup time by **30%**.

FINAL-YEAR INTERNSHIP – INDUSTRIAL AND ENVIRONMENTAL RISKS
OCP – Safi Apr. – Jun. 2024
- Analyzed and ranked risks associated with PS4 project changes using the ADRPT method.
- Assessed environmental impacts to identify improvement priorities and preventive measures.

FINAL-YEAR INTERNSHIP – PROCESS DIAGNOSTICS AND INDUSTRIAL QUALITY
OCP – Safi Apr. – Jun. 2023
- Investigated faults on freshwater treatment lines and supported corrective work to improve treatment quality and cycle stability.

TECHNICAL SKILLS

Product requirements and integration: Functional analysis, mechanical and electrical interfaces, supplier specification review, comparison of design alternatives and V-model.
Product configuration and change control: Bills of materials (BOM), revision tracking, PLM traceability, and basic knowledge of ECR/ECN/ECO and product release.
Quality and industrialization: Risk and root cause analysis (5 Whys, Ishikawa, Pareto); basic knowledge of APQP, PPAP, DFMEA, PFMEA, 8D and SPC.
Verification and validation: Test preparation, MIL simulation, fault injection and analysis of results; basic knowledge of DVP&R and SIL.
Lean and industrial methods: 5S, Kaizen, PDCA, value stream mapping, waste identification, cycle time, takt time and OEE.
Tools and reporting: CATIA V5, Cameo (SysML), MATLAB/Simulink–Stateflow, Aras Innovator (basic), Excel, Power BI and MS Project.
Embedded systems: C/C++, Python, Git, STM32, Raspberry Pi, CAN, sensors and actuators.

ACADEMIC PROJECTS

- **Automotive suspension design:** designed and modeled suspension components in CATIA V5.
- **Multi-ECU automotive system:** used C/C++ on STM32F446RE and Raspberry Pi 5 connected over CAN; implemented 4WD, ACC, FCW and AEB functions using MATLAB/Simulink–Stateflow, with Qt monitoring and MIL/SIL validation through fault injection.
- **Embedded vision and ADAS:** developed driver drowsiness detection and automatic emergency braking functions using YOLOv8, OpenCV, MediaPipe and ESP32.
- **Regenerative braking:** modeled a control strategy for recovering braking energy in MATLAB/Simulink.

EDUCATION

National School of Applied Sciences (ENSA) – Berrechid 2024 – Present
Engineering degree program in Automotive and Digital Solutions, final year.
Higher School of Technology (EST) – Safi 2021 – 2024
Professional Bachelor's Degree in Metrology, Quality, Safety and Environment.
University Diploma of Technology (DUT) in Instrumental Techniques and Quality Management.
High School Diploma – Experimental Sciences 2020 – 2021

LANGUAGES AND CERTIFICATIONS

Arabic: Native **French:** B2 **English:** B1

Modeling and Simulation with Simulink – MathWorks (Coursera), May 2026.
Automobile Engineering: How Modern Cars Are Built – Starweaver.